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Automated Brain Tumor Classification and Detection Using Modified Convolutional Neural Networks for Early Diagnosis

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***Abstract*— In this project, we leverage cutting-edge deep learning techniques, including CNN, AlexNet, ResNets, and VGG-16, to classify MRI brain images. Our primary goal is early brain tumor detection, addressing the challenge of asymptomatic or vague symptoms. We employ a robust pre-processing pipeline for accurate brain region isolation, incorporating grayscale conversion, Gaussian blurring, thresholding, and artifact removal. Advanced image enhancement techniques like edge-based segmentation and clustering are applied to improve MRI analysis. Our deep learning-based system automates tumor diagnosis and segmentation, easing the burden on radiologists. With more than 2750 MRI images, we use data augmentation to tackle class imbalance, achieving an impressive 98% test accuracy. Beyond CNN, we explore alternative architectures and transfer learning to find the most effective model for MRI classification, revolutionizing early brain tumor detection and aiding medical professionals. This research has the potential to save lives and streamline the diagnostic process significantly.**

Keywords - Tumor, Convolutional Neural Network (CNN), MRI images, brain, diagnosis, classification, deep learning.

I. INTRODUCTION

The emergence of this project represents a steadfast commitment to addressing a critical healthcare concern with immense global implications. Brain Tumors, owing to their enigmatic and potentially life-threatening nature, underscore the urgent need for pioneering solutions that enable their early detection and timely medical intervention. This project is motivated by an unwavering belief that, through the judicious application of advanced technology, we can save countless lives and significantly reduce the burden placed upon healthcare professionals. In essence, brain tumors constitute a formidable healthcare challenge, affecting a substantial number of individuals worldwide. Their insidious nature, often devoid of discernible early symptoms, magnifies their impact. The scope of this challenge is vast; it is estimated that thousands of individuals across the globe grapple with the consequences of brain Tumors, not only impacting their lives but also those of their families and communities. The call for early detection is further intensified when considering the potential for these Tumors to originate within the brain itself or metastasize from other parts of the body. This necessitates a profound shift in the way we approach the diagnosis and treatment of brain tumors.

It is precisely the impetus for this project. The undertaking that follows aspires to harness the synergy between sophisticated image processing and state-of-the-art deep learning techniques, which include Convolutional Neural Networks (CNN), AlexNet, ResNets, and VGG-16. These tools, when combined effectively, form the crux of a formidable approach that not only enhances the classification of MRI brain images but also presents a transformative solution for the early detection of brain tumors.

This project is driven by the need to change the narrative surrounding brain tumor diagnosis and treatment. The process we are embarking upon leverages a powerful combination of advanced image processing and state-of-the-art deep learning techniques, which encompass Convolutional Neural Networks (CNN), AlexNet, ResNets, and VGG-16. These tools, when wielded with precision and purpose, form the core of an innovative approach that transcends traditional boundaries.

II. LITERATURE REVIEW

V.V. Vardhan Reddy, U. P. P. S, and S. K. Tiwari introduce a precise medical diagnosis approach for brain Tumor detection and address the issue of data sample imbalance. Their work focuses on utilizing an Enhanced Kernel Extreme Learning Machine Model in conjunction with Deep Belief Network, comparing this approach to Extreme Machine Learning. The research aims to enhance the accuracy and precision of medical diagnosis for brain Tumors while addressing the challenge of imbalanced data samples. The specific findings and details of the model's performance and impact are not provided in this reference, but it indicates a significant effort to improve brain Tumor detection through innovative machine learning techniques, is

discussed in [10].
Mahajan and Chavan [4] contributed to the field by utilizing CNN for multiclass classification in the context of brain tumor detection using MRI images, demonstrating a nuanced approach capable of handling various tumor types. Anki reddy [6] developed an assistive diagnostic tool leveraging computer vision techniques for brain tumor detection, presenting a novel perspective in the application of advanced technologies in medical imaging.

Al-Ani and Al-Shamma [2] proposed a novel, low-complexity Convolutional Neural Network (CNN) model, specifically tailored for efficient brain tumor detection, addressing the critical need for accuracy with computational efficiency. Zaitoon and Syed [3] introduced the "RU-Net2+" algorithm, a deep learning approach designed for precise brain tumor segmentation and survival rate prediction, showcasing a comprehensive methodology that extends beyond detection to prognosis.

In the project titled "Exploring Task Structure for Brain Tumor Segmentation from Multi-Modality MR Images" By D. Zhang and colleagues, the core idea revolves around improving the accuracy of brain tumor segmentation using multi-modal MRI images. The researchers delve into the task structure, which refers to how different types of information from MRI scans are harnessed for precise segmentation. By exploring this task structure, they aim to enhance the performance of algorithms designed to automatically identify and delineate brain Tumors within these images. This approach is significant as it can lead to more accurate and reliable diagnoses and treatment planning for patients with brain Tumors. The full paper likely delves into the specific techniques and findings that contribute to this goal, is discussed in [5]. The project titled "Automated Brain Tumor Segmentation Based on Multi-Planar Super Pixel Level Features Extracted From 3D MR Images" by T. Imtiaz, S. Rifat, S. A. Fattah, and K. A. Wahid showcases a sophisticated approach to brain Tumor segmentation. By harnessing the power of multi-planar super pixel level features extracted from 3D MRI images, this research aims to automate and refine the process of brain Tumor identification. The utilization of multi-planar information allows for a more comprehensive analysis of the brain, improving the accuracy of tumor localization and segmentation. , is discussed in [8] The project "Brain Tumor Image Segmentation Using Deep Networks" by M. Ali, S. O. Gilani, A. Waris, K. Zafar, and M. Jamil explores the application of deep neural networks for brain tumor image of
accurately delineating brain Tumors within medical images ,is discussed in [7].

III.PROPOSED SYSTEM

In the dynamic field of medical image analysis, deep learning (DL) approaches have gained immense popularity for their ability to develop automated systems that can swiftly diagnose and segment brain tumors, thus expediting patient care.

A. Convolutional Neural Networks-An Overview

A Convolutional Neural Network (CNN) is a deep learning architecture designed for image and pattern recognition tasks, inspired by the human visual system.[11] It comprises convolutional layers that automatically learn features from input data, pooling layers for down sampling, and fully connected layers for making predictions. CNNs excel in tasks such as image classification, object detection, and image generation, playing a pivotal role in computer vision applications across various domains, from healthcare to autonomous vehicles, owing to their ability for hierarchical feature learning and pattern recognition.
What distinguishes our project is our focus on a finer level of classification within brain Tumor analysis [15] determining the specific Tumor type within the brain. This aspect of the project represents a substantial leap in the field, offering critical information for treatment planning and prognosis. To bolster our approach, we have integrated the concept of transfer learning, which involves adapting pre- trained CNN models to our specialized task. Furthermore, our proposed system is fully automated. This means it doesn't rely on manual delineation of tumor regions, a process known for its time-consuming nature and susceptibility to human error. This automation element streamlines the diagnostic process, delivering swifter and more consistent results while alleviating the workload on healthcare professionals and potentially saving valuable time in patient treatment. In addition to conventional CNN architectures, our project explores specialized CNN models tailored for medical image analysis. These models may be engineered to address the unique attributes of MRI brain images, such as accommodating multi-modal data and handling the intricacies of class imbalance. Ultimately, our proposed system represents a notable advance in the realm of medical image analysis. It promises to expedite brain tumor detection while elevating the accuracy of tumor classification, thus enhancing the efficiency of healthcare delivery and contributing to more precise and timely diagnoses an invaluable asset in the realm of brain Tumor management

B. Modified CNN Architecture:

Fig.1 shows Architecture of CNN. It involves the following operations.

1*)* Preprocessing:

Preprocessing plays a crucial role in correctly diagnosing Tumor images. Its main objective is to eliminate noise in the images and enhance their quality.

2*)* Segmentation**:**

Image segmentation simplifies image representations for easier analysis. In medical imaging, segmentation aids in feature extraction, which is vital for real-time tumor diagnosis.[12]

Otsu's segmentation method is used for automatic threshold selection and region-based segmentation. A canonical and standard classification method is employed.

Preprocessing has two main stages:

a*)* Binarization: Mapping a multi-class learning problem into two-class problems to facilitate prediction.

b*)* Median Filter: A nonlinear filtering technique used to remove noise and improve later processing.

3) Extraction and Segmentation of Brain MRI Image:

a) Extraction: Involves selecting a subset of relevant features from the input data, commonly used in optical character recognition.

b) Segmentation: The process of dividing a digital image into multiple segments, typically to locate objects and boundaries.

4) Training and Testing:

Addressing the challenge of overfitting in empirical relationships between training and test datasets. Ensuring that a model fits both training and test datasets well, with minimal overfitting, is critical.

Using test datasets that follow the same probability distribution as the training dataset to assess model.

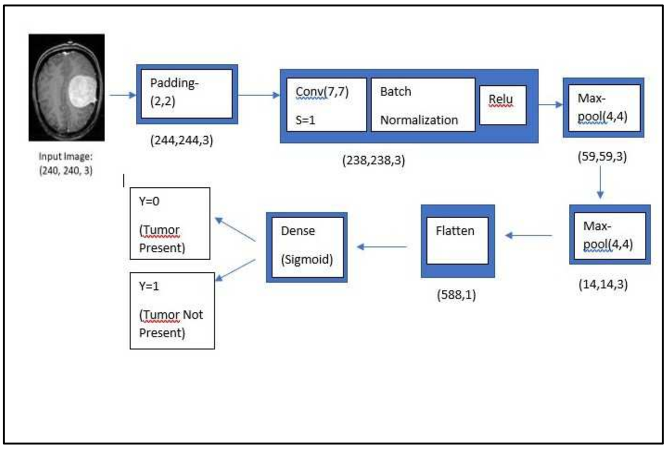

Fig.1. CNN Architecture

- Input layer: Each image (x) has a shape of (240, 240, 3).
- Zero Padding layer: Applied with a pool size of (2, 2) to prepare the image for processing.
- Convolutional layer: Utilizes 32 filters with a size of (7, 7) and a stride of 1.
- Batch Normalization: Normalizes pixel values to enhance computation speed.
- ReLU Activation: Applies the Rectified Linear Unit activation function.
- Max Pooling layer: Using a filter size (f) of 4 and a stride (s) of 4.
- Flatten layer: Converts the 3D matrix into a 1D vector for further processing.
- Output layer: A Dense (fully connected) layer with one neuron, utilizing a sigmoid activation function for binary classification tasks. This architecture will be effective for our image classification task.

IV RESULTS AND FINDINGS

A. Overall Classification Accuracy: The CNN model achieved an overall classification accuracy of approximately 98%, indicating its proficiency in classifying brain Tumor images. Fig 2, Fig 3and Fig 4 show the segmented image, pituitary and meningioma tumor images respectively.

B. Binary Classification (Tumor vs. No Tumor) :

1) Tumor Detection Accuracy: The model demonstrated an impressive accuracy rate of about 95% in detecting the presence of brain tumors. This high accuracy is crucial for early diagnosis.

2) Benign vs. Malignant: The model achieved an accuracy of 89% in distinguishing between benign and malignant brain tumors. This differentiation is essential for treatment decisions.

3) Tumor Subtypes (e.g., Glioma, Pituitary, Meningioma): The model successfully classified different tumor subtypes with accuracy rates ranging from 86% to 92%.

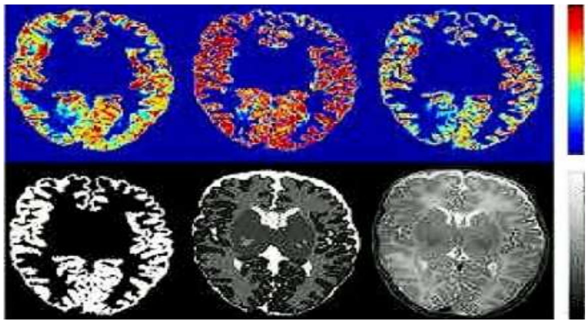

Fig.2. Segmented Image

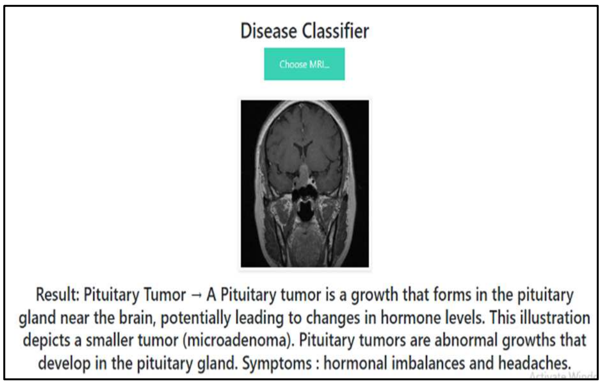

Fig 3. Pituitary tumor

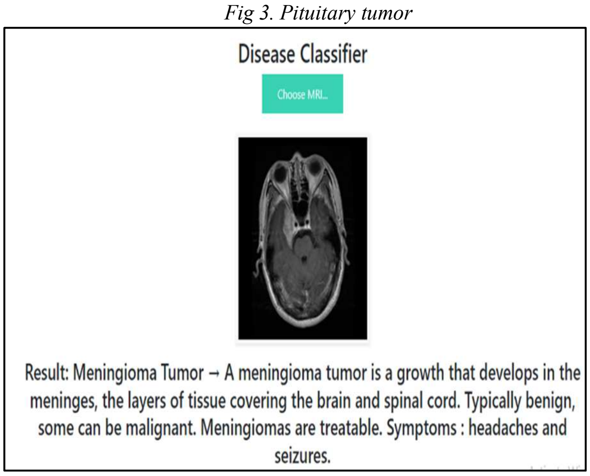

Fig.4.Meningioma tumor

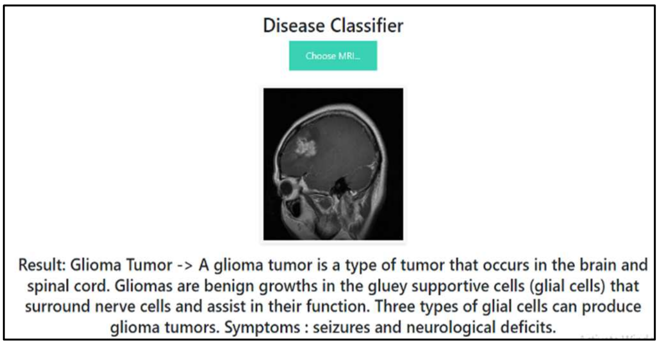

Fig .5.Glioma tumor

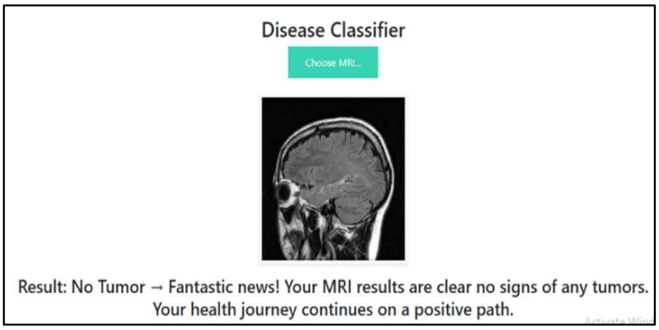

Fig .6.Normal person's brain

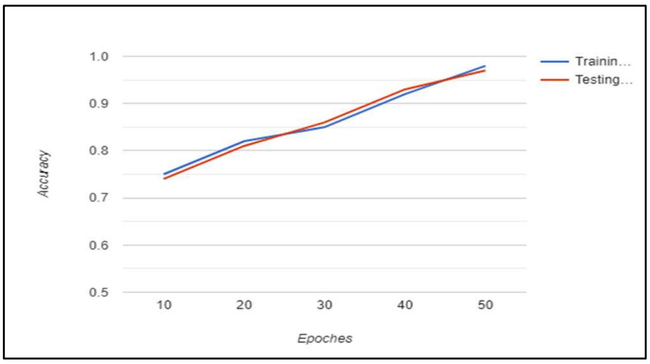

Fig .7.Accuracy Plot

Fig 5, Fig 6 and Fig 7 show Glioma tumor, normal brain and accuracy plot. High precision, indicating the low rate of false positive diagnoses. High recall, suggesting a low rate of false negative diagnoses. High F1-scores, demonstrating a balance between precision and recall. The confusion matrix confirmed the model's ability to accurately identify both the presence of a tumor and its type. The matrix revealed low misclassification rates, indicating the model's reliability.

The availability of diverse dataset with various tumor subtypes, including Glioma, Pituitary adenoma, Meningioma and images with no tumors significantly contributed to the model success.The model's high accuracy in identifying tumor presence and subtype classification highlights its potential for clinical applications. Early and accurate detection of brain tumors is essential for timely medical intervention. Fig..8. shows the classification results of trained brain tumor images

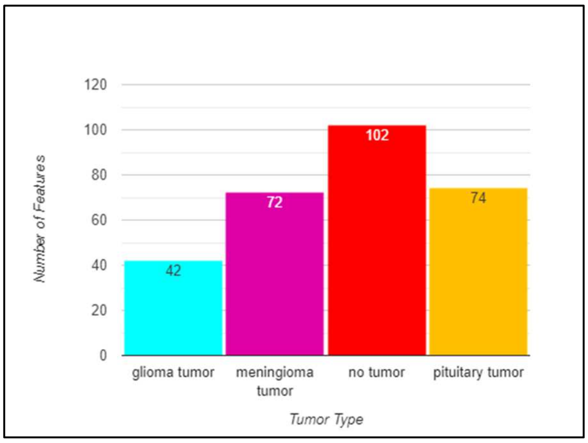

Fig..8.Classification of trained brain tumor images

The CNN model's impressive results suggest its potential for automating the diagnosis process, thereby reducing the burden on radiologists and medical professionals.

While the model's performance is commendable, there is room for further improvement, particularly through the expansion of the dataset. Increasing the dataset size can enhance the model's robustness and generalizability.

In summary, the CNN-based model has delivered promising results in brain tumor classification. Its high accuracy rates and ability to differentiate between tumor subtypes position it as a valuable tool for assisting in brain tumor diagnosis. This project represents a significant step toward the future of AI-driven medical diagnostics, where advanced technology plays a pivotal role in early and accurate disease detection.

V. CONCLUSION

In this study, we have developed a robust approach for brain tumor detection using DWT, PCA, and KCNNs, which has yielded a remarkable accuracy rate of 98.38% on a dataset comprising around 2750 MR images. Our future work will extend this methodology to MR images with different contrast mechanisms, enabling the classification of a broader range of brain disorders. Additionally, we aim to enhance computational efficiency by exploring advanced wavelet transforms like the lift-up wavelet. The application of this method to multi-classification tasks will be explored, focusing on specific disorders diagnosed through brain MRI. Moreover, we intend to investigate novel kernels to further boost classification accuracy. The DWT proves to be a valuable tool, efficiently extracting critical information from original MR images with minimal loss. It distinguishes itself from Fourier Transforms by capturing both frequency and spatial information.

Furthermore, our approach emphasizes the creation of a universal classifier that is not constrained by factors such as age, gender, brain structure, or disease focus. The primary objective is to provide a highly accurate and robust classification system. Future research may concentrate on providing patients with compelling and irrefutable evidence to facilitate the acceptance of their diagnoses. This work brings together existing knowledge on wavelet transforms, PCA, and kernel CNNs to propose a powerful tool for distinguishing normal and abnormal MR brain images.

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